

Generation Interconnection System Impact Study Report

For Queue Project Q15-0553

Alta View 230 kV

January 28, 2018

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q15-0553. The Interconnection Customer proposes a 475.6 MW wind facility interconnecting at the Alta View 230 kV substation in the Kentucky area. The request was received on December 16, 2015. The study evaluated system intact and single contingency conditions using the regional planning case, including 188 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 1 overloaded facility. The total estimated cost of the network upgrades allocated to this project is \$23,882,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: Ala View 345 kV/230 kV XFMR
Contingency: Ata View to Whitten 230 KV
Loading: 129.3% (Rating: 368 MVA)
DFAX: 100.0%
Pre-project loading: 0.0%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Ala View 345 kV/230 kV XFMR	Replace transformer with higher-rated unit	\$23,882,000	100% (\$23,882,000)
	Total allocated to project		\$23,882,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.